

Chapter 8

Water Flow in Capillary Tube

8.1 Objectives

1. Learn to apply and interpret graphs.
2. Determining the half-life of the exponential decay function in the case of water flow in a capillary tube, for various diameters and lengths.

8.2 Tools

- 1 burette with stopcock on a stand
- 1 measuring beaker (100 ml)
- 1 graduated cylinder (50 ml)
- 1 stopwatch
- 1 vernier capiler

8.3 Theoretical Foundation

A capillary tube is connected to a burette with a stopcock, and the water level in the burette is measured using a ruler. If the height of water in the burette (h), the decrease in water height (Δh), the time interval of flow (Δt), then the decrease in water level is proportional to the flow time interval:

$$\Delta h \sim \Delta t \quad (8.1)$$

The water flow rate (Q) through the capillary tube in laminar flow satisfies (Poiseuille's Law),

$$Q = \frac{\Delta V}{\Delta t} = \frac{\pi r^4 \Delta P}{8\eta L} \quad (8.2)$$

Noted:

- ΔV = volume of water flowing through the tube during the time interval Δt
- r = radius of the capillary tube (half of the hole diameter)
- ΔP = pressure difference between the two ends of the capillary tube
- η = viscosity of water
- L = length of the capillary tube

For the water flow in the burette: $Q = A \frac{\Delta h}{\Delta t}$ or $\Delta h = \frac{Q}{A} \Delta t$

Where (A) is the cross-sectional area of the burette. By substituting the value of Q from equation (8.2), we obtain:

$$\Delta h = \frac{\pi r^4 \Delta P}{8A\eta L} \Delta t \quad (8.3)$$

The pressure difference between the two ends of the capillary tube (ΔP) is proportional to the water height in the burette (h), which satisfies:

$$\Delta P = \rho g h \quad (8.4)$$

with, ρ = the density of water, and g = the acceleration of gravity. From equations (8.3) and (8.4), we obtain:

$$\Delta h = \frac{\pi r^4 \rho g h}{8A\eta L} \Delta t \quad (8.5)$$

or

$$dh = -\lambda h dt \quad (8.6)$$

with λ = a constant is referred to as the decay constant, the negative sign (–) indicates a decrease in pressure during the flow process. The equation (8.5) is a differential equation with the solution given by:

$$h = h_0 e^{-\lambda t} \quad (8.7)$$

with values

$$\lambda = \frac{\pi r^4 \rho g}{8A\eta L} \quad (8.8)$$

where: h_0 = the height of the water surface in the burette at time ($t = 0$), t = the flow time.

8.4 Experimental Methods

8.4.1 Apparatus Schematic

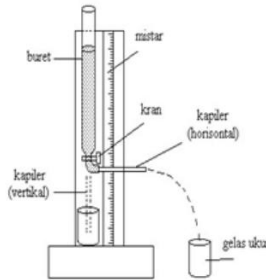


Figure 8.1: Schematic diagram of apparatus

Tool Functions:

- Ruler: indicates the height level of the water (h)

- Stopcock: to open/close the flow
- Measuring Cylinder serves as the water container.

The experiment is conducted with the capillary tube positioned both horizontally and vertically, also by varying the length and diameter of the capillary opening. *(Discuss this with your lab assistant)!*

8.4.2 Observation Procedure

1. Fill the burette with water and measure the initial water surface height (h_0), which is the height measured from the tip of the capillary tube. Then, measure the water surface height (h) at each time interval of 10 seconds.
2. Repeat the procedure for various lengths of capillary tubes (L) with the same diameter (0,05 mm), as well as for capillary tubes of the same length but with different diameters, namely 0,5 mm and 0,3 mm.
3. Perform data collection as described above, with horizontal and vertical positions of the capillary tube.

8.4.3 Observation Data Table

Data Table 1

(Various pipe lengths, Horizontal position)

No	t (second)	h (cm)		
		L_1 = 5 cm	L_2 = 10 cm	L_3 = 15 cm
1	0	$h_0 = \dots$	$h_0 = \dots$	$h_0 = \dots$
2	10			
3	20			
4	30			
....				

Data Table 2

(Various pipe lengths, Vertical position)

No	t (second)	$h \text{ (cm)}$		
		L_1 $= 5 \text{ cm}$	L_2 $= 10 \text{ cm}$	L_3 $= 15 \text{ cm}$
1	0	$h_0 = \dots$	$h_0 = \dots$	$h_0 = \dots$
2	10			
3	20			
4	30			
....				

Data Table 3

(Various pipe diameters, Horizontal position)

No	t (second)	$h \text{ (cm)}$		
		d_1 $= 0,3 \text{ mm}$	d_2 $= 0,5 \text{ mm}$	d_3 $= 1,0 \text{ mm}$
1	0	$h_0 = \dots$	$h_0 = \dots$	$h_0 = \dots$
2	10			
3	20			
4	30			
....				

Data Table 4

(Various pipe diameters, Vertical position)

No	t (second)	h (cm)		
		d_1 = 0,3 mm	d_2 = 0,5 mm	d_3 = 1,0 mm
1	0	$h_0 = \dots$	$h_0 = \dots$	$h_0 = \dots$
2	10			
3	20			
4	30			
....				

8.4.4 Analysis Method

According to the equation $h = h_0 e^{-\lambda t}$ it becomes $\ln h = \ln h_0 - \lambda t$

1. From each data table, plot a graph of $\ln h$ vs t , from the analysis of the graph, determine the decay constant (λ), which corresponds to the slope of the graph.
2. Determine the half-life ($t_{1/2}$) from obtained value of λ . (What is meant by half-life? Find a reference!)
3. Given that the density of water at room temperature is 1 g/cm^3 , determine the viscosity (η) of water under various conditions using your observation data.
4. Compare your experimental results with the reference values provided in literature source (2).
5. Draw conclusions from the discussion of the experimental results you have obtained.

(**Note:** all analysis must be accompanied by their respective uncertainties)

8.4.5 Schematic of the Analytical Graph Model

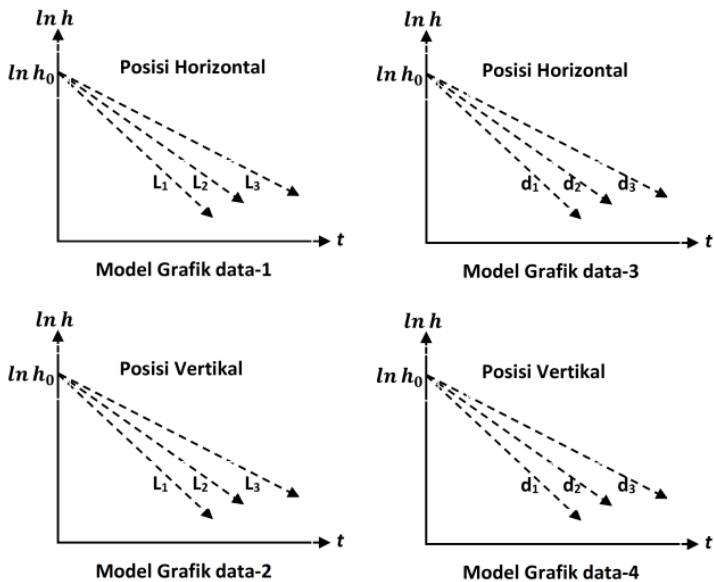


Figure 8.2

8.5 Reference

1. W Sears: Mechanics, Heat and Sound.
2. Hand Book of Chemistry and Physics, 60th edition 11979-1980CRC. Press, page F.51.